



# High-throughput and rapid classification on harmful algal bloom species based on mega image database and artificial intelligence

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## ABSTRACT

Microalgae are essential components of marine ecosystems and have significant industrial applications. However, their rapid identification, especially HAB species, poses a challenge. This study constructed a comprehensive microalgae image database and developed AI-based classification algorithms to improve identification accuracy. Using DenseNet, EfficientNet, and ViT models, we achieved high classification performance, with ViT showing the best results. The study highlighted the importance of dataset size and diversity in enhancing model performance. Additionally, the trained model was applied to analyze field-collected samples, and the results were compared with those obtained from microscopic examination and metabarcoding analyses. The application of AI technology in microalgae classification provides a reliable basis for early warning and rapid response to HABs, reducing ecological and economic losses. Our findings demonstrate the potential of AI in enhancing the speed and accuracy of microalgae identification, contributing to better management and protection of marine ecosystems.

## 1. Introduction

Microalgae play a crucial role in marine ecosystems as foundational organisms that support various ecological functions and industries (Soyocak et al., 2023; Wu et al., 2021). As primary producers, microalgae convert sunlight into energy through photosynthesis, forming the base of the marine food web (Bagul et al., 2024). They contribute to carbon fixation, help reduce greenhouse gas emissions, and improve water quality through nutrient cycling (Sharma and Arya, 2023). The commercial development of microalgae has expanded into fields such as biofuels, pharmaceuticals, and nutraceuticals, highlighting their economic significance (Dineshkumar et al., 2024). However, the rapid reproduction or aggregation of algae can lead to harmful algal blooms (HABs) (Yu et al., 2018), which not only impact water quality and ecological balance but also pose significant risks to fisheries, tourism, and public health (Berdalet et al., 2016; Kouakou and Poder, 2019). The Gulf of Mexico is heavily influenced by anthropogenic activities, and HABs pose significant threats to both the local economy and the marine

environment. Recurrent blooms of *Karenia brevis* are of particular concern, as this species produces neurotoxic shellfish poisoning (NSP) toxins, which pose serious risks to human health (Baden, 1989; Hardman, 2003; Soto et al., 2012; Ulloa et al., 2017). Similarly, the coastal marine ecosystems of China have undergone profound changes over the past five decades, characterized by significant alterations in nutrient concentrations and composition in seawater. These changes have driven shifts in microalgal community structures, with an increasing frequency of toxic microalgal blooms and the emergence of multiple newly recorded bloom-forming species (Dai et al., 2014; Gu et al., 2013; Lu et al., 2014; Wang et al., 2011; Zhou et al., 2008). Therefore, the swift and accurate identification of microalgae, particularly HAB species, is essential (Anderson, 2009; Lang et al., 2024; Li et al., 2025; Zahir et al., 2024).

There are estimated to be between 30,000 and 1 million species of phytoplankton in freshwater and marine ecosystems worldwide, with around 8000 species of diatoms and 2200 species of dinoflagellates named, cataloged, and described (Guiry, 2012; Rachman et al., 2022).

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Currently, the classification and identification of these phytoplankton primarily rely on microscopic examination, a traditional method that is time-consuming and labor-intensive, making it challenging for an individual to recognize and differentiate such a diverse range of microalgae (Cui et al., 2021; Pok and Ahmad, 2024). To address this issue, various methods have been employed for detecting microalgae in environmental samples, including molecular technologies like high-throughput sequencing and quantitative real-time PCR (qPCR) (He et al., 2023; Hu et al., 2024; Liu et al., 2020). The Imaging FlowCytobot (IFCB), developed by the Woods Hole Oceanographic Institution (WHOI), combines imaging and flow cytometry techniques to rapidly capture images of microalgae in environmental samples, enhancing the efficiency of sample analysis (Olson and Sosik, 2007). This device has been used to detect phytoplankton in the Gulf of Mexico, the Baltic Sea, and Antarctic waters, playing a crucial role in the early warning of HAB species (Henrichs et al., 2021; Kraft et al., 2021; Lee et al., 2022). The large volume of image data generated by IFCB necessitates automated processing using image classification algorithms (Sosik and Olson, 2007). Deep learning algorithms, such as convolutional neural networks (CNNs), automatically learn image features without the need for expert extraction, reducing human involvement and achieving excellent performance in image classification (Li et al., 2022). Rachman et al. (2022) used CNN models to automatically identify genus-level phytoplankton in Indonesian waters with an accuracy of over 75 %. Fuchs et al. (2022) combined convolutional neural networks with automated pulse-shape recording flow cytometry in their study. Eerola et al. (2020) used a compact CNN architecture to classify phytoplankton communities in the Baltic Sea, employing effective data augmentation methods to achieve high classification accuracy. Several studies have employed optical microscopy to acquire images of microalgae, which were then manually annotated to facilitate automated detection of microalgal species (Sun et al., 2025b; Zhou et al., 2023). Lang et al. (2024) utilized a three-dimensional imaging-based approach to enable the automated detection of harmful algal species.

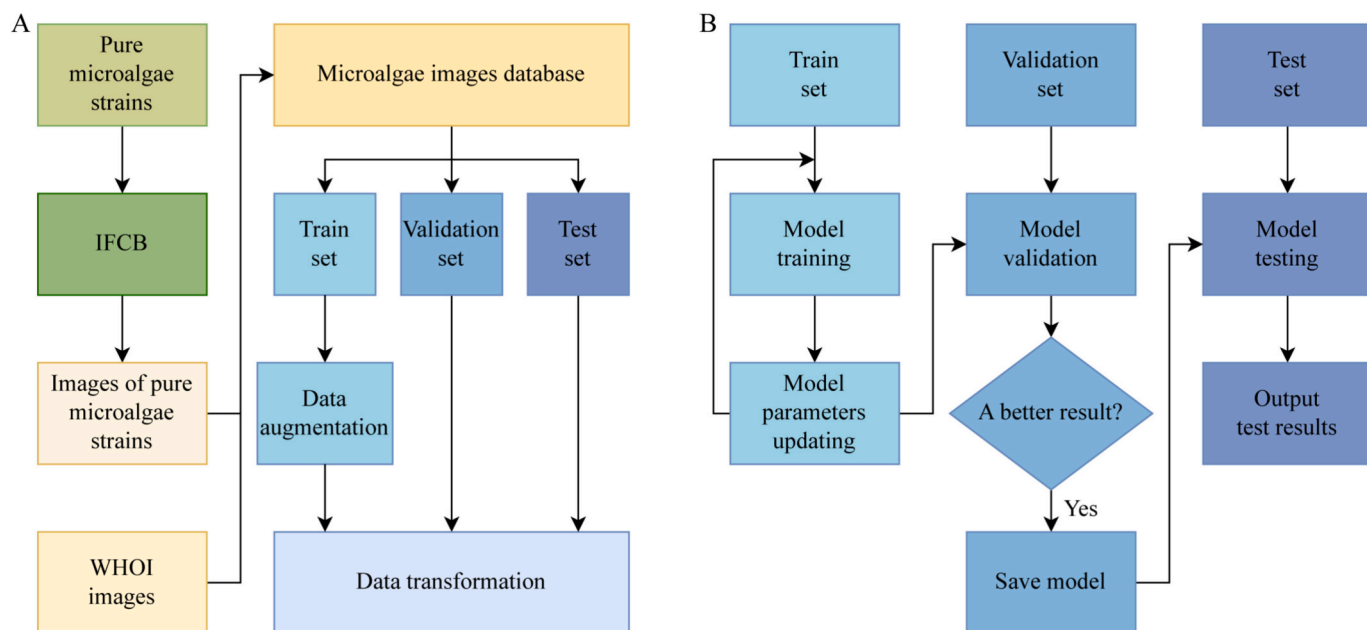
In this study, we used IFCB to collect microalgae images, constructing an image database encompassing various microalgae species. We developed and compared multiple artificial intelligence-based classification algorithms. We also applied the models to seawater samples from the Beibu Gulf and compared the classification results with those

obtained from microscopic and metabarcoding analyses. This process aims to enhance the identification accuracy of microalgae, particularly HAB species, providing critical support for early warning and further microalgae automatic classification research.

## 2. Materials and methods

### 2.1. Construction of microalgae image database

In this study, we constructed a microalgae images database, involving the collection and processing of microalgae images (Fig. 1A). The Microalgae Group at the Second Institute of Oceanography, Ministry of Natural Resources, has built a repository of over 800 pure microalgae strains through years of field sampling, laboratory isolation, purification, and long-term cultivation. A subset of these strains have been taxonomically identified to the genus and species level. We selected 112 strains of microalgae from different genera and species, and acquired their images using an Imaging FlowCytobot (IFCB, McLane Research Laboratories, Inc., United States). After removing impurities and debris, images of the same category were merged, and for each category, more than 1000 images were randomly selected. If a category had fewer than 1000 images, all images in that category were included. The final pure microalgae strains image library comprises 92 microalgae species and 86,495 images. To further expand the diversity of the dataset, we incorporated microalgae images from the publicly available IFCB plankton images dataset of the Woods Hole Oceanographic Institution (WHOI) (Sosik et al., 2015). This dataset includes phytoplankton, zooplankton, and other categories such as detritus and bubble. Among these, there are 59 categories of phytoplankton, mostly classified to the genus level. We merged images of various zooplankton categories into a single “Zooplankton” category and combined detritus, bubble, etc. into an “Other” category. Subsequently, we merged the pure microalgae strains image library with the WHOI images dataset to form a comprehensive microalgae images database. Each category in the database was created by randomly selecting 1000 images. If a category contained fewer than 1000 images, all images from that category were included, while ensuring that the proportion of images from different sources within the same category was as balanced as possible. Ultimately, the database comprises 142 categories (of which 140 are microalgae),



**Fig. 1.** Microalgae image database construction and deep learning model training process. (A) Microalgae images collection and database construction process; (B) Model training, validation and testing process.

encompassing a total of 106,606 images. To support the subsequent training of deep learning models, we divided the image data into training, validation, and test sets in a ratio of 60 %, 20 %, and 20 %, respectively. This division ensures the model's generalization capability during training and allows for effective evaluation of the model's performance on unseen data (Muraina, 2022). Additionally, we applied various data augmentation techniques to the training set, such as random cropping and random flipping, to enhance the diversity of the dataset and improve the model's generalization capability. The validation and test sets underwent image resizing and center cropping. All three datasets were subject to data transformation processes, including converting single-channel grayscale images to three-channel grayscale images, converting image arrays to tensors, and normalizing images to meet the input requirements of different deep learning models.

To investigate the relationship between training set size and model recognition capability, we extracted 90 categories from the database, each containing 1000 images, and created seven smaller datasets. The validation and test sets for these datasets were kept consistent, while each category in the training sets contained 600, 300, 100, 50, 20, 10, and 2 images, respectively. And each smaller training set is randomly extracted from the previous larger training set.

## 2.2. Construction of image classification model

In this study, we selected three deep learning models: DenseNet (Densely Connected Convolutional Networks), EfficientNet, and Vision Transformer (ViT). These models have demonstrated exceptional performance in image classification tasks. DenseNet, introduced by Huang et al. in 2017 at the Computer Vision and Pattern Recognition (CVPR) conference, is a model based on convolutional neural network (CNN) architecture. The core idea of DenseNet is to connect each layer directly with each subsequent layer. This dense connection structure greatly improves the feature propagation, alleviates the vanishing gradient problem, improves the model's anti-overfitting ability, at the same time reduces the number of parameters and computational cost. DenseNet outperformed previous models on the CIFAR (Krizhevsky and Hinton, 2009) and Street View House Numbers (SVHN) datasets (Netzer et al., 2011), and achieved comparable results on the ImageNet dataset (Deng et al., 2009) with fewer parameters (Huang et al., 2017). DenseNet was therefore awarded the Best Paper Award at CVPR 2017. EfficientNet, proposed by Tan and Le in 2019 at the International Conference on Machine Learning (ICML), is also a model based on CNN architecture. The researchers designed a standardized CNN scaling method, known as compound scaling, to simultaneously scale the model's width, depth, and resolution in proportion, utilizing Neural Architecture Search (NAS) to determine the model's structure. EfficientNet balances network performance and computational cost, achieving higher accuracy with fewer parameters and lower computational requirements. It surpassed all previous models on the ImageNet dataset and achieved similar performance on other datasets with fewer parameters (Tan and Le, 2019). EfficientNet's compound scaling strategy reduces unnecessary computation, resulting in a well-balanced performance and efficiency. ViT, introduced by Dosovitskiy et al. in 2021 at the International Conference on Learning Representations (ICLR), is a model based on the Transformer architecture. The Transformer model, relying solely on attention mechanisms, has revolutionized the natural language processing (NLP) field since it was first proposed, with the renowned GPT (Generative Pre-trained Transformer) also based on this architecture (Radford et al., 2018; Vaswani et al., 2017). Instead of modifying the Transformer architecture, ViT treats images as sequences, transforming image processing tasks into sequence modeling tasks, and the fine-tuned ViT outperforms the best CNN models of the time on datasets such as ImageNet (Dosovitskiy et al., 2021). To adapt to the microalgae image classification tasks in this study, we used the parameters of the three models after pre-training with ImageNet and adjusted the output layer size while keeping their overall frameworks unchanged.

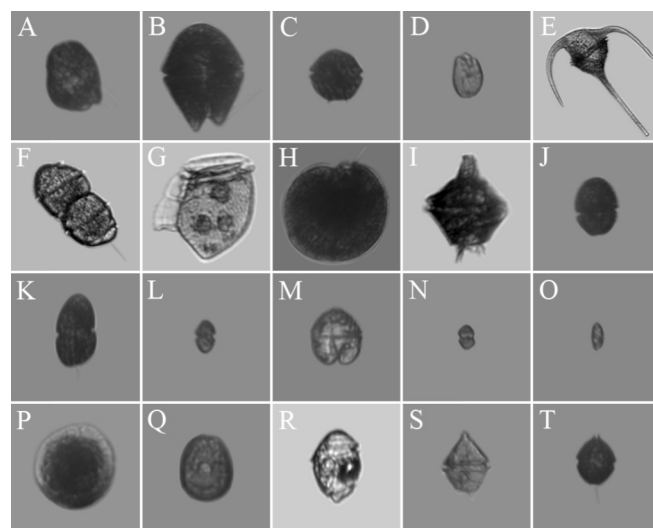
## 2.3. Model training and testing

In this study, we employed the PyTorch deep learning framework (Paszke et al., 2019) to implement the training, validation, and testing of the microalgae images classification model (Fig. 1B). These processes were conducted on a workstation equipped with an NVIDIA A100 GPU at the Fourth Institute of Oceanography, Ministry of Natural Resources China, to ensure efficient computational performance. We utilized a combination of grid search and manual selection to set the model's hyperparameters, including batch size, learning rate and epoch. Once the hyperparameters were set, the model began iterative updates, requiring 32 or 64 training epochs per iteration, with each epoch comprising training and validation phases. During the training phase, the model performed gradient calculations and backpropagation, updating parameters using the cross-entropy loss function and stochastic gradient descent (SGD) optimizer. In the validation phase, the model only performed forward propagation to evaluate performance, without backpropagation and parameter updates. After every 7 training epochs, the learning rate scheduler was used to adjust the learning rate, enhancing the training effectiveness. Throughout the training and validation processes, we calculated loss, accuracy, precision, recall, and F1 score. The F1 score represents the weighted average of precision and recall, effectively balancing both metrics. Hence, we utilized F1 score as the critical indicator for assessing model performance. If the F1 score in the validation phase surpassed the previous best result, the current model parameters were saved. Finally, the model's performance was evaluated on the test set.

## 3. Result

### 3.1. Microalgae image database distribution

To apply artificial intelligence to the rapid detection of microalgae, this study constructed an image database comprising 140 categories of microalgae (annotated to genus or species level), including 28 potential HAB species. Additionally, the database includes two other categories: Zooplankton and non-phytoplankton/non-zooplankton objects (Other), resulting in a total of 142 categories. Table S1 shows the details of the



**Fig. 2.** Dinoflagellate plates. A: *Adenoides sinensis*; B: *Akashiwo sanguinea*; C: *Alexandrium pacificum*; D: *Amphidinium trulla*; E: *Ceratium* spp.; F: *Cochlodinium* spp.; G: *Dinophysis* spp.; H: *Gambierdiscus caribaeus*; I: *Gonyaulax* spp.; J: *Gymnodinium dorsalisulcum*; K: *Gyrodinium* sp.2; L: *Heterocapsa niei*; M: *Karenia mikimotoi*; N: *Karlodinium* sp.1; O: *Moestrupia* sp.1; P: *Ostreopsis* sp.1; Q: *Procerentrum lima*; R: *Proterothropsis* spp.; S: *Protoperidinium* spp.; T: *Scrippsiella trochoidea*. (E, F, G, I, R, S: Images from WHOI's images dataset).

number of images in each category. Figs. 2 and 3 respectively display examples of dinoflagellates, diatoms, and other microalgae, showcasing the wide range of species and morphological diversity within the dataset. Among these, many are HAB species found along the Chinese coast, such as *Akashiwo sanguinea* (Fig. 2B), *Alexandrium pacificum* (Fig. 2C), and *Karenia mikimotoi* (Fig. 2M), which are common harmful dinoflagellates. Some species within the genus *Cochlodinium* have been reclassified into the genus *Margalefidinium*. However, since the WHOI image library was constructed earlier and did not make this distinction, we have continued to use the original classification in our study. Diatoms include *Asterionellopsis* (Fig. 3A), *Chaetoceros didymus* (Fig. 3C), and *Guinardia delicatula* (Fig. 3K), other microalgae such as *Chattonella* sp.1 (Fig. 3P) and *Heterosigma akashiwo* (Fig. 3S) are also common HAB species.

The distribution characteristics of each category within the microalgae image database reveals significant imbalance. Specifically, 90 categories contain 1000 images, while nearly 40 % of the categories have fewer than 1000 images (Fig. 4A). By dividing the data into training, validation, and test sets with a ratio of 60 %, 20 %, and 20 %, respectively, the number of images per category in the training set ranges from 1 to 600, while in the validation and test sets, the number of images per category ranges from 1 to 200. There are 35 categories with 100–1000 images, primarily ranging from 258 to 666 images (Fig. 4D); 13 categories with 10–100 images, with a relatively dispersed distribution (Fig. 4C); and 4 categories with fewer than 10 images. Of these, one category has only 4 images, with one image randomly selected for the validation and test sets each, and the remaining two images used for the training set. Another category contains only 2 images, with one image randomly selected for the training set and the remaining image shared by the validation and test sets (Fig. 4B).

### 3.2. Model training and validation process

To evaluate the performance of deep learning models in the task of microalgae image classification, this study selected three models: DenseNet, EfficientNet, and ViT. Various hyperparameters were set for training, validation, and testing, and the best-performing models on the

test set were ultimately selected (as detailed in Table S2). Fig. 5 shows the trends of loss and F1 scores calculated for these three models on the training and validation sets over the training epochs. These observations show that the training process for the three models is very similar. In the early stage, the model rapidly converges, gradually flattens out after about 10 epochs, and remains stable with only slight fluctuations in the later stage. And the fluctuation on the validation set is smaller than that on the training set. The loss on both the training and validation sets was relatively low, and the F1 scores were high. The models performed slightly better on the validation set compared to the training set, indicating that there was no underfitting or overfitting. All three models effectively captured the classification features of the data and demonstrated good generalization ability. The performance differences among the three models were minimal, with F1 scores on the validation set exceeding 0.9, and the ViT model showing slightly better performance than the other two CNN models.

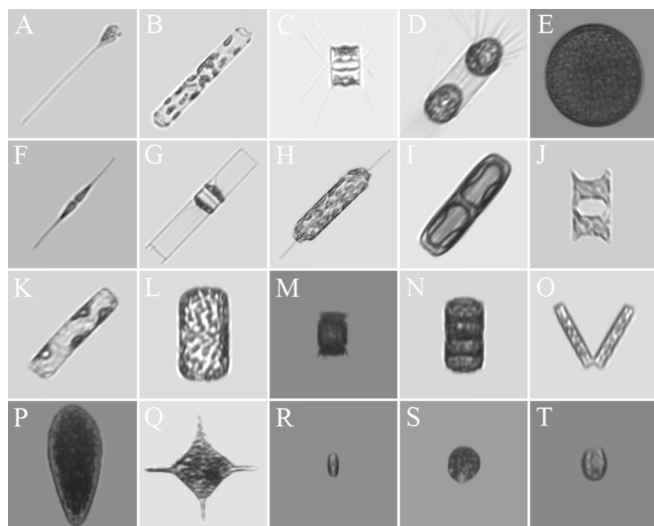
### 3.3. Model testing and evaluation results

After completing a full training process, we obtained the optimal parameters for the current model on the validation set and subsequently applied this model to the test set for evaluation. We selected the best-performing DenseNet, EfficientNet, and ViT models on the test set (Table S2). The test results showed that the best-performing DenseNet had an F1 score of 0.9048 on the test set, the best EfficientNet had an F1 score of 0.9169, and the best ViT had an F1 score of 0.9205. Fig. 5 presents the F1 score distribution for these models across different categories in the test set. Overall, the three models exhibited high classification performance on the test set, with F1 scores exceeding 0.9 for the entire set. Particularly for categories with a large sample size, the models provided more accurate classifications (100–1000 group and 1000 group). However, for categories with 10–100 images, all models showed significant performance fluctuations. As we can see, DenseNet's F1 score mainly ranging between 0 and 0.8182 (Fig. 5D), and EfficientNet's F1 score mainly ranging between 0.4 and 0.8667 (Fig. 5E), F1 score of ViT performed better than the other models, mainly ranging between 0.5 and 0.9333 (Fig. 5F). And for categories with fewer than 10 images, the F1 scores were almost all 0. This indicates that sample size significantly impacts the performance of deep learning models, and categories with fewer samples pose a greater challenge for these models.

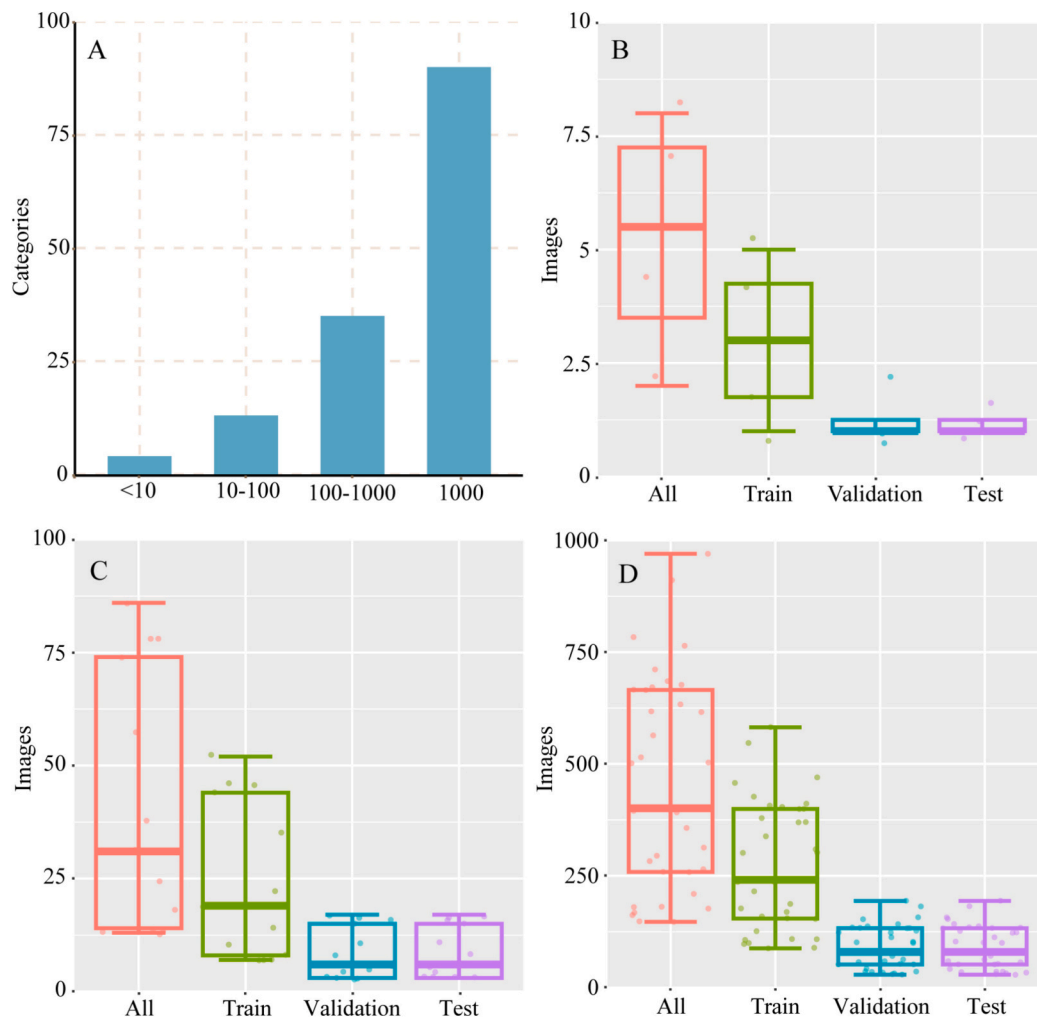
## 4. Discussion

### 4.1. Overall model performance

Traditional classification and identification of microalgae samples rely on microscopic examination, which is time-consuming and labor-intensive, and heavily dependent on individual expertise (Ning et al., 2022; Sun et al., 2025a). Automatic classification of microalgae images is a solution to this problem (Chong et al., 2024). Traditional machine learning algorithms require manual feature extraction, necessitating expertise from various domains, whereas deep learning algorithms can automatically learn features, further reducing manual effort (Chauhan and Singh, 2018; Sosik and Olson, 2007). Deep learning algorithms have matured in the field of image classification and have shown excellent performance (Jiao and Zhao, 2019). DenseNet, EfficientNet, and ViT, three deep learning models, have demonstrated good classification performance on datasets like ImageNet (Dosovitskiy et al., 2021; Huang et al., 2017; Tan and Le, 2019). We applied these models to the microalgae classification domain, and they performed well on our dataset. The overall F1 scores of the three models on the test set exceeded 0.9, with ViT achieving the highest score of 0.9205, followed by EfficientNet at 0.9169, and DenseNet at 0.9048. This ranking is consistent with their performance on ImageNet, where ViT-based networks rank first, EfficientNet-based networks second, and DenseNet-based networks third (Yu et al., 2022; Pham et al., 2021; Kim et al., 2025). There is a



**Fig. 3.** Plates of diatoms and other microalgae. A: *Asterionellopsis* spp.; B: *Cerataulina* spp.; C: *Chaetoceros didymus*; D: *Corethron* spp.; E: *Coscinodiscus jonesianus*; F: *Cylindrotheca* spp.; G: *Dactyliosolen* spp.; H: *Ditylum parasite*; I: *Ephemia* spp.; J: *Eucampia* spp.; K: *Guinardia delicatula*; L: *Lauderia* spp.; M: *Odontella aurita*; N: *Paralia* spp.; O: *Thalassionema* spp.; P: *Chattonella* sp.1; Q: *Dictyocha* spp.; R: *Dunaliella salina*; S: *Heterosigma akashiwo*; T: *Olisthodiscus* sp.1. (A–O: Diatoms, P–T: Other microalgae; A, B, C, D, F, G, H, I, J, K, L, N, O, Q: Images from WHOI images dataset).



**Fig. 4.** Distribution of microalgae image database. A: histogram of the overall situation of the database; B: Distribution of <10 group; C: Distribution of 10–100 group; D: Distribution of 100–1000 group (<10 group: Categories containing fewer than 10 images; 10–100 group: Categories containing the number of images between 10–100; 100–1000 group: Categories containing the number of images between 100–1000; 1000 group: Categories containing 1000 images).

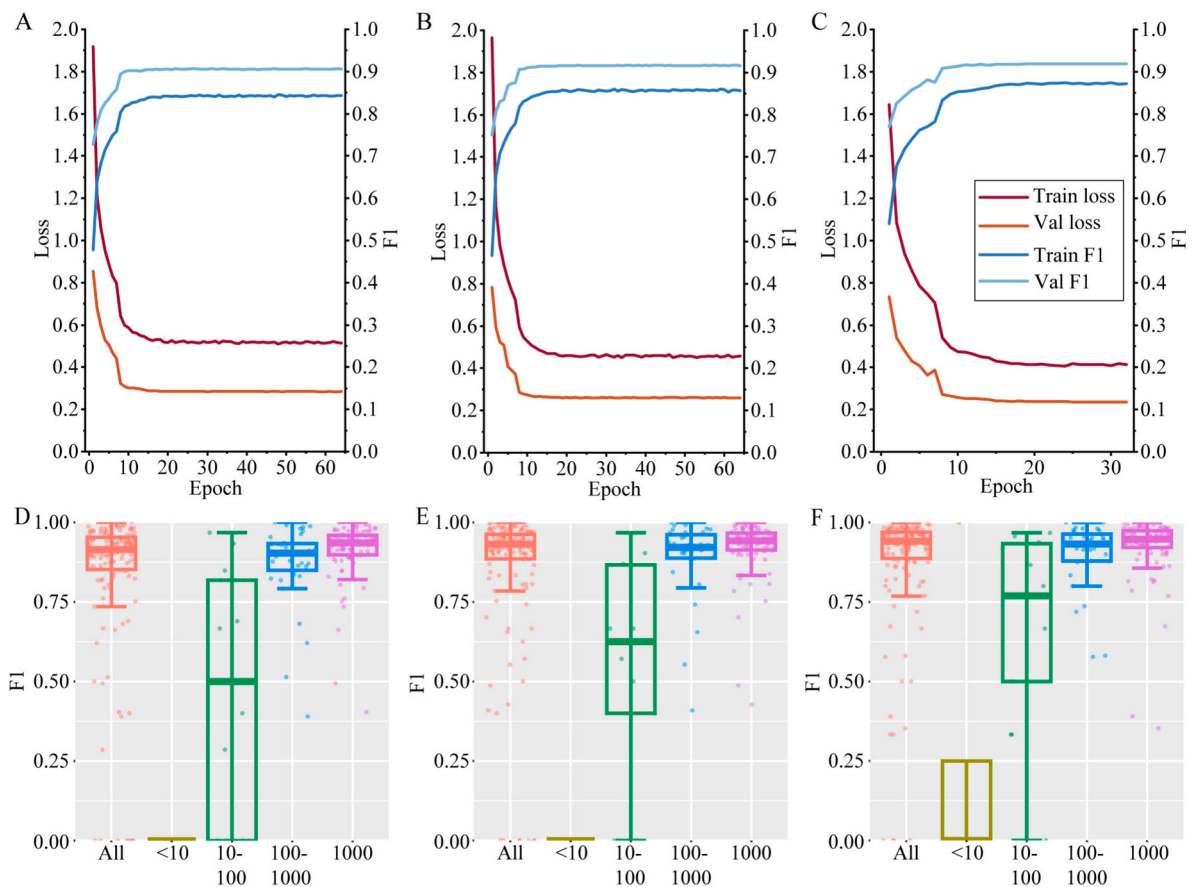
significant relationship between model performance on the dataset and the number of model parameters (Ong et al., 2023). Among the three models mentioned, DenseNet has over 20 million parameters, EfficientNet has over 54 million, and ViT has over 86 million. However, larger models have a higher risk of overfitting, which can degrade model performance and generalization (Gupta et al., 2018). Regularization methods such as dropout (Hinton et al., 2012) and batch normalization (Ioffe and Szegedy, 2015) were applied during model training. As shown in Fig. 5, the model achieved higher F1 scores on the validation set compared to the training set, indicating good generalization performance, which is related to regularization and data augmentation during training. Additionally, models like DenseNet and EfficientNet, with fewer parameters, require less training and inference time, making them more suitable for deployment on hardware-constrained mobile platforms (Chang et al., 2023). It can be observed that each training epoch involves 63,958 images. On our computing setup, DenseNet requires 189 s per epoch, EfficientNet requires 215 s, and ViT requires 27 min. For inference, DenseNet takes 1.83 s to process 1000 images, EfficientNet takes 1.78 s, and ViT takes 9.24 s.

Beyond model performance, the quality of the dataset plays a crucial role in achieving robust classification. Many existing microalgae image datasets include only a few to a dozen categories, with total image counts rarely exceeding a few hundred or thousand. In contrast, our dataset encompasses a substantially larger number of categories and images, providing a clear advantage for training deep learning models

and improving classification performance. Additionally, our dataset was collected using IFCB, whereas most other datasets rely on traditional optical microscopy. While some microscopy-based datasets allow direct annotation of both microalgae categories and their positions within images, our dataset focuses exclusively on identifying microalgae (Sun et al., 2025b; Zhou et al., 2023). Nevertheless, the use of IFCB enables rapid processing of seawater samples, significantly enhancing throughput and operational efficiency.

#### 4.2. Model performance on different categories

The three models achieved different F1 scores across various categories, with the top 10 categories shown in Table S3. Most categories with high F1 scores contain a large number of images in the dataset, with seven categories appearing in the top 10 of all three models' F1 scores, indicating that the unique structures of certain microalgae species enhance model classification performance. Although *Emiliania huxleyi* has fewer samples, its F1 scores are high across all three models, likely due to its unique shape features (Paasche, 2001). The bottom 10 categories with the lowest F1 scores, as shown in Table S4, include six categories that appear in the lowest 10 of all three models, with DenseNet and EfficientNet both scoring 0 for these categories. Most categories with low F1 scores contain fewer images in the dataset, further highlighting the impact of sample quantity on deep learning model performance (Fang et al., 2021; Shahinfar et al., 2020). *Ditylum parasite*,



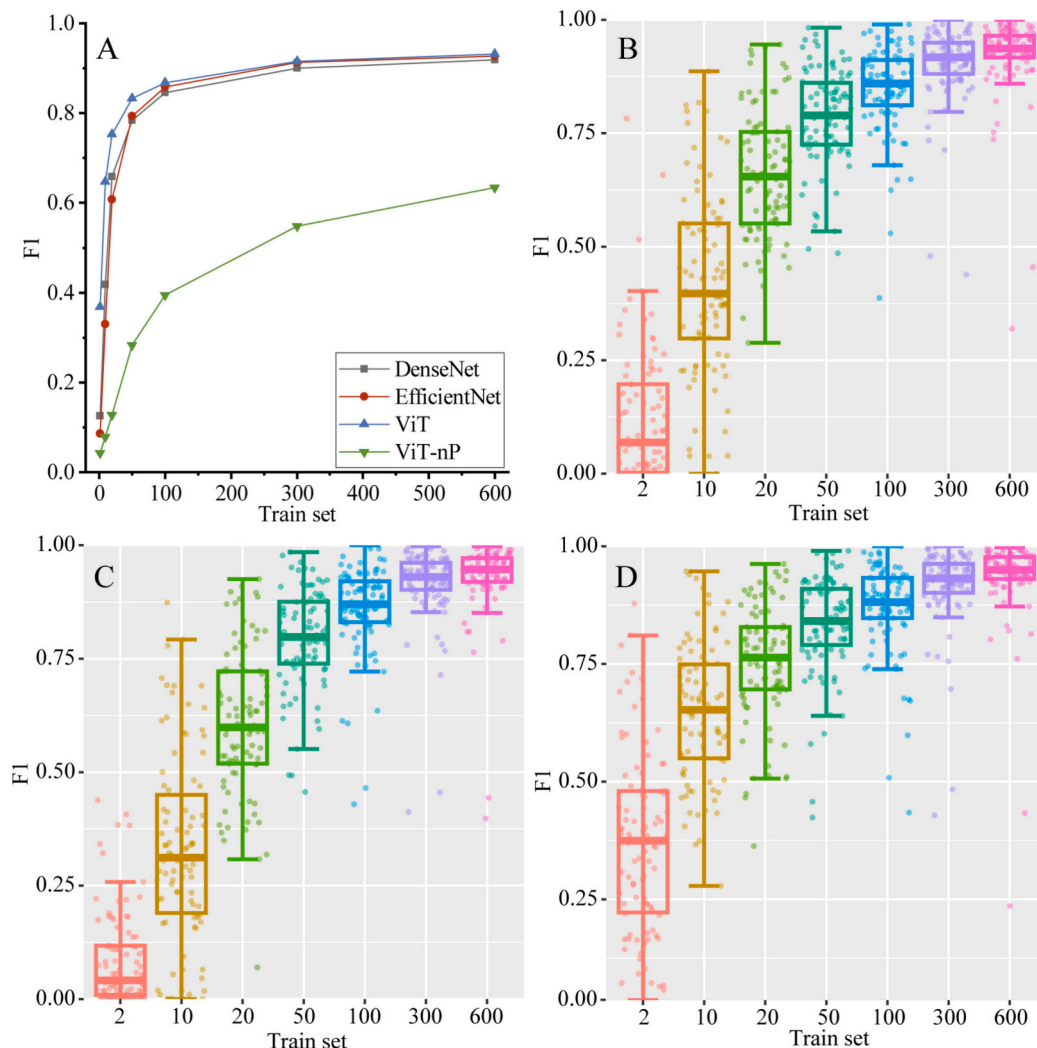
**Fig. 5.** Variation trend of loss and F1 score on train set and validation set during model training. (A: The training process of DenseNet; B: The training process of EfficientNet; C: The training process of ViT.) Performance of the models for each category across different groups within the test set. (D: F1 score distribution of DenseNet on the test set; E: F1 score distribution of EfficientNet on the test set; F: F1 score distribution of ViT on the test set. All group: All categories; <10 group: Categories containing fewer than 10 images; 10–100 group: Categories containing the number of images between 10–100; 100–1000 group: Categories containing the number of images between 100–1000; 1000 group: Categories containing 1000 images.)

*Karlodinium* sp.1, and *Karlodinium* spp. have a relatively large number of images, especially *Karlodinium* sp.1 and *Karlodinium* spp., but their classification accuracy remains low. Tables S5, S6, and S7 show that *D. parasite* is often misclassified as *Ditylum* spp., and *Karlodinium* sp.1 and *Karlodinium* spp. are mutually misclassified. This may be because some images of *D. parasite* are also present in *Ditylum* spp., and some data of *Karlodinium* sp.1 are indeed included in *Karlodinium* spp. This indicates that the models can extract features for analysis and clustering, and even have some ability to correct the database (Rottmann and Reese, 2023; Xie et al., 2018). *Coolia palmyrensis* is often misclassified as *Coolia tropicalis* (Tables S5, S6, S7), suggesting that these classification models may recognize similarities within the genus. The models exhibit similar classification errors, such as *Amphidinium tomasii* being classified as *Togula* sp.1, *Hemiaulus* spp. as *Ditylum* spp., and *Gambierdiscus caribaeus* as *Ostreopsis* sp.1, indicating that the models may detect morphological similarities among these microalgae species, providing insights for microalgae system evolution, new species identification, and discovery. Tables S5, S6, and S7 also show that for these high-error-rate categories, DenseNet disperses them across more categories, totaling 35 categories, while EfficientNet and ViT each have 26, with ViT correctly classifying *Alexandrium tamarense*, consistent with the F1 score rankings of the three models. DenseNet has the lowest overall F1 score, and the higher the classification error dispersion, the worse the model performance, which can be used as an indicator of model quality (Pal et al., 2012). The top 10 categories with the largest F1 score differences among the three models, shown in Table S8, generally have low recognition rates and fewer samples, indicating that the features of these categories

have not been well learned, resulting in higher classification randomness (Khobragade et al., 2022; Rubio-Manzano, 2024). Among these categories, ViT generally performs best, with *A. tamarense* having an F1 score of 1 in ViT, while DenseNet and EfficientNet score 0. Similarly, *Cochlodinium* spp. and *Odontella* spp. have F1 scores over 0.8 in ViT but lower in other models. Although these categories have fewer than 500 images, their F1 scores are relatively high in all models, with an average sample size of around 250, which is significantly larger than categories with fewer than 100 images.

#### 4.3. Relationship between dataset size and model performance

To verify the relationship between training set size and model recognition ability, datasets of different sizes were created, using the same validation and test sets for model training and performance evaluation. As shown in Fig. 6A, all models improved their classification performance as the training data increased, particularly from 2 to 50 images per category, where the improvement rate was very rapid. However, from 300 to 600 images, the improvement rate was much slower. Regardless of the dataset size, the ViT model consistently achieved the highest F1 scores, especially when each category had fewer than 20 images. ViT's classification performance was significantly higher than the other models, while EfficientNet only surpassed DenseNet when the training set for each category exceeded 50 images (Fig. 6A–D). It is noteworthy that the ViT model without pre-training performed poorly on the test set across all training set sizes (Fig. 6A), yet its improvement trend was still apparent. This might be because the



**Fig. 6.** Performance of models trained on different-sized train sets. Tested on the same test set. A: Overall performance of each model on the test set after training on different-sized train sets; B: F1 score distribution of DenseNet on the test set after training on different-sized train sets; C: F1 score distribution of EfficientNet on the test set after training on different-sized train sets; D: F1 score distribution of ViT on the test set after training on different-sized train sets. (X-axis represents the number of images per category in the train set; ViT-nP: ViT model without pre-training on other large datasets.)

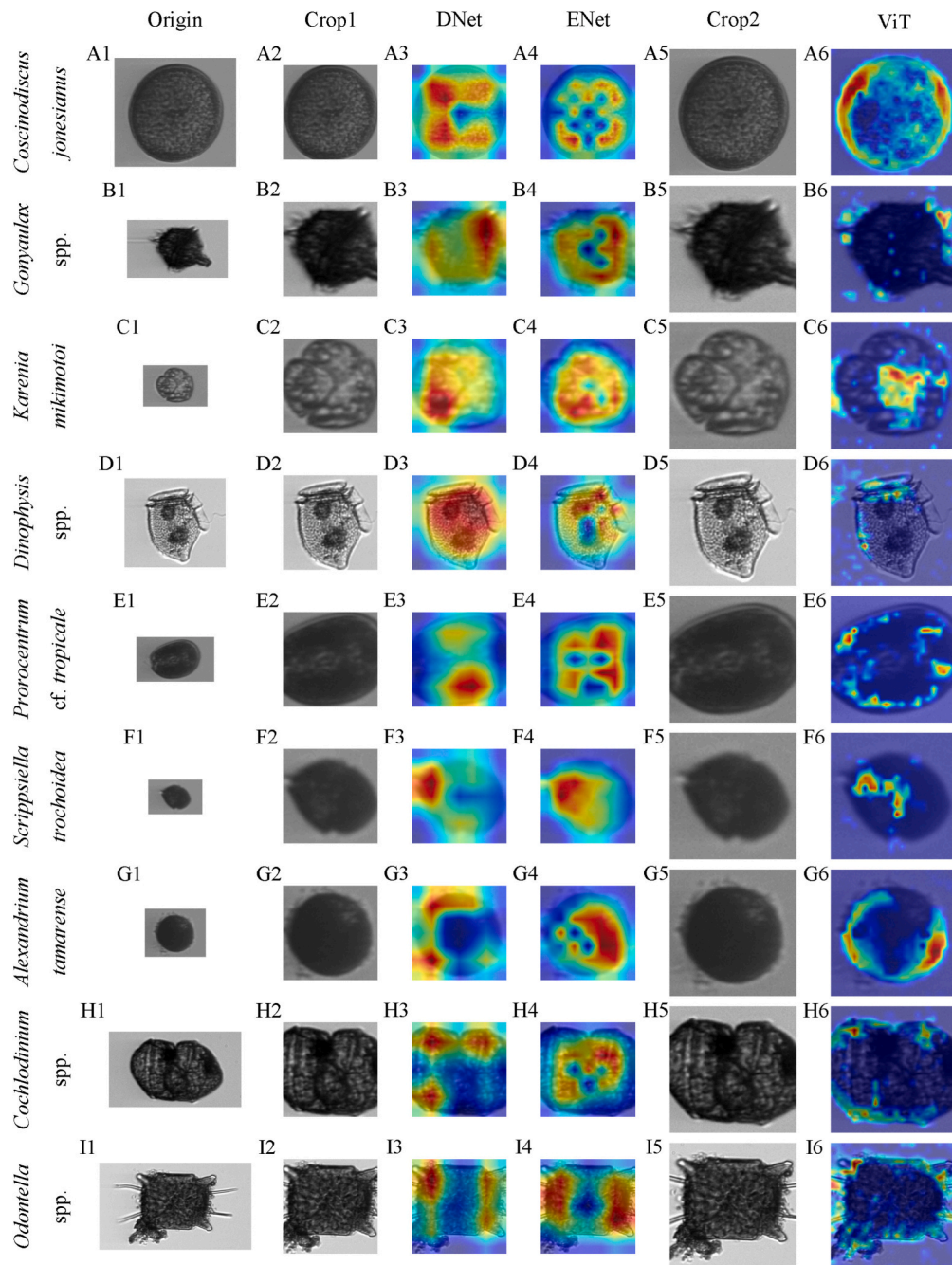
ViT model requires a large amount of data for training, and a pre-trained ViT model performs well even with smaller datasets (Dosovitskiy et al., 2021; Wu et al., 2023). Fig. 6 clearly illustrates the relationship between data volume and model classification ability. The development of deep learning is heavily reliant on the accumulation of large amounts of data. ImageNet plays a crucial role in image classification because high-quality, accurately annotated data is essential for this task (Deng et al., 2009; Emam et al., 2021; Truong et al., 2022; Zhou et al., 2017). The microalgae images collected from our laboratory are more species-specific compared to those from the WHOI database, which are often classified only to the genus level. The pure culture and isolation of microalgae, and the establishment of a physical algal strain library, are significant for research and information collection (Afiya and Umar, 2022; Singh et al., 2015). Microalgae exhibit different morphologies at various growth stages or under different growth conditions, which also need to be considered (Cortés-Altamirano and Sierra-Beltrán, 2003; Falasco et al., 2009).

Currently, the microalgae image database is not mature enough to form a sufficiently large and precise dataset. Improving the construction of the microalgae image database would undoubtedly play a crucial role in the rapid identification of microalgae, especially HAB species. According to our experiments, collecting more than 300 images per

category for training yields better results (Fig. 6). Using the best-performing ViT model as an example, we demonstrated that some categories achieved high F1 scores even with a small number of training images (Table S10). These categories could be well distinguished from others even after training with just two images. Conversely, Table S11 shows categories with low F1 scores despite having large training sets. The poor classification performance of *Karlodinium* sp.1 and *Karlodinium* spp., as discussed earlier, is due to the overlap in images between the two categories, leading to mutual misclassification. For *Leptocylindrus* spp., *Rhizosolenia* spp., and *Pseudonitzschia* spp., their similar morphology leads to mutual misclassification, reducing their F1 scores (Table S12).

#### 4.4. Model visualization explanation

To understand how models classify microalgae images and the differences between them, we used the Gradient-weighted Class Activation Mapping (Grad-CAM) method proposed by Selvaraju et al. (2017), as shown in Figs. 7 and 8. In the selected images, *Coscinodiscus jonesianus*, *Gonyaulax* spp., and *Karenia mikimotoi* are categories with high F1 scores in DenseNet, EfficientNet, and ViT (Fig. 7A–C). *Dinophysis* spp., *Procerocentrum* cf. *tropicale*, and *Scrippsiella trochoidea* are categories with



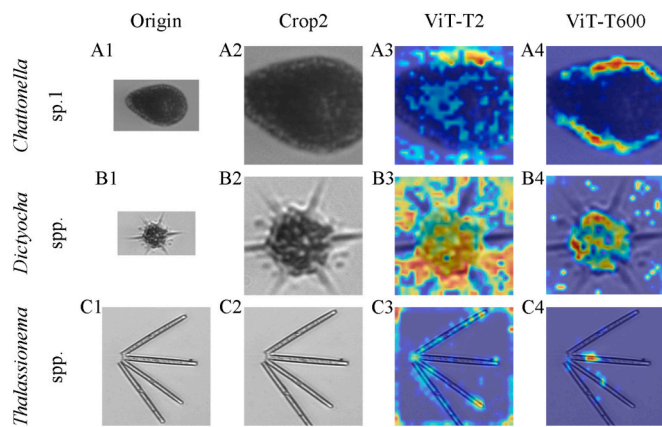
**Fig. 7.** Examples of Gradient-weighted Class Activation Maps (Grad-CAM) for different models. (Origin: Original image; Crop1: Transformed and cropped image used as input for DenseNet and EfficientNet; DNet: DenseNet; ENet: EfficientNet; Crop2: Transformed and cropped image used as input for ViT.)

high F1 scores even with fewer images in all models (Fig. 7D–F). *Alexandrium tamarense*, *Cochlodinium* spp., and *Odontella* spp. are categories with significant F1 score differences among the three models (Fig. 7G–I). From Fig. 7, we can see that each model focuses on the morphology of the microalgae, which has the most significant impact on classification. DenseNet and EfficientNet, both CNN-based models, have multiple focus areas within the microalgae, possibly emphasizing texture (Heinert et al., 2024). ViT has fewer focus areas, mainly concentrating on the edges of the microalgae, which might be related to its self-attention mechanism, allowing it to find distant feature correlations (Dosovitskiy et al., 2021). Fig. 8 shows categories with high F1 scores despite small training sets. However, there are still significant differences between models trained on only two images and those trained on 600 images. Although both can correctly identify the categories, the

focus areas are more scattered with fewer training images. Deep learning models can learn data features autonomously, but their interpretability has always been a challenge, especially for complex models like ViT (Chakraborty et al., 2017; Wu et al., 2023). Better understanding of model mechanisms in the future will allow for more effective training and application of artificial intelligence (Ennab and Mcheick, 2024; Linardatos et al., 2020).

#### 4.5. AI classification of potential HAB species

The three models used in this study demonstrated good classification performance for potential HAB species, especially the ViT model. Except for *Gambierdiscus caribaeus*, which had an F1 score of only 0.5, the ViT model achieved F1 scores above 0.9 for the other 27 species (Table S13).



**Fig. 8.** Examples of Gradient-weighted Class Activation Maps (Grad-CAM) for ViT models trained on different-sized train sets. (Origin: Original image; Crop2: Transformed and cropped image used as input for ViT; ViT-T2: Train set with 2 images per category; ViT-T600: Train set with 600 images per category.)

This highlights the excellent performance of AI in identifying harmful algal bloom species and its significant role in the warning of HABs. *Akashiwo sanguinea* is widely distributed in global estuaries and coastal waters, posing significant threats to the environment and aquatic organisms. Its toxicity has acute lethal effects on various aquatic animals, such as fish, shellfish, and shrimp, and is enhanced under eutrophic conditions (Koening et al., 2014; Xu et al., 2017). In Yantai Sishili Bay, China, *A. sanguinea* has caused annual HABs, and it once dramatically decreased scallop yields by more than 170 hm<sup>2</sup>. This species has high adaptability to low-nutrient environments and can form resting cysts, posing a continuous threat to ecosystem stability and coastal economic activities (Chen et al., 2015; Wu et al., 2001). *Heterosigma akashiwo* is another HAB species, causing massive fish mortality and economic losses. HABs can persist for weeks to months and frequently occur in coastal regions worldwide. Climate change and global warming may increase the frequency and severity of HABs (Mehdizadeh Allaf, 2023). Its toxicity mechanisms include neurotoxins, cytotoxins, and reactive oxygen species, which can cause fish gill blockage. HABs not only harm fish and aquatic ecosystems but also cause hypoxia in bottom waters (Giesler et al., 2023). In Canada, the United States, Japan, and New Zealand, *H. akashiwo* blooms have caused severe impacts and economic losses (Mehdizadeh Allaf and Trick, 2019; Powers et al., 2012). The application of AI technology can significantly enhance the speed and accuracy of identifying potential HAB species, providing a reliable basis for rapid response and early warning, thereby reducing ecological and economic

losses.

#### 4.6. Field validation using environmental samples

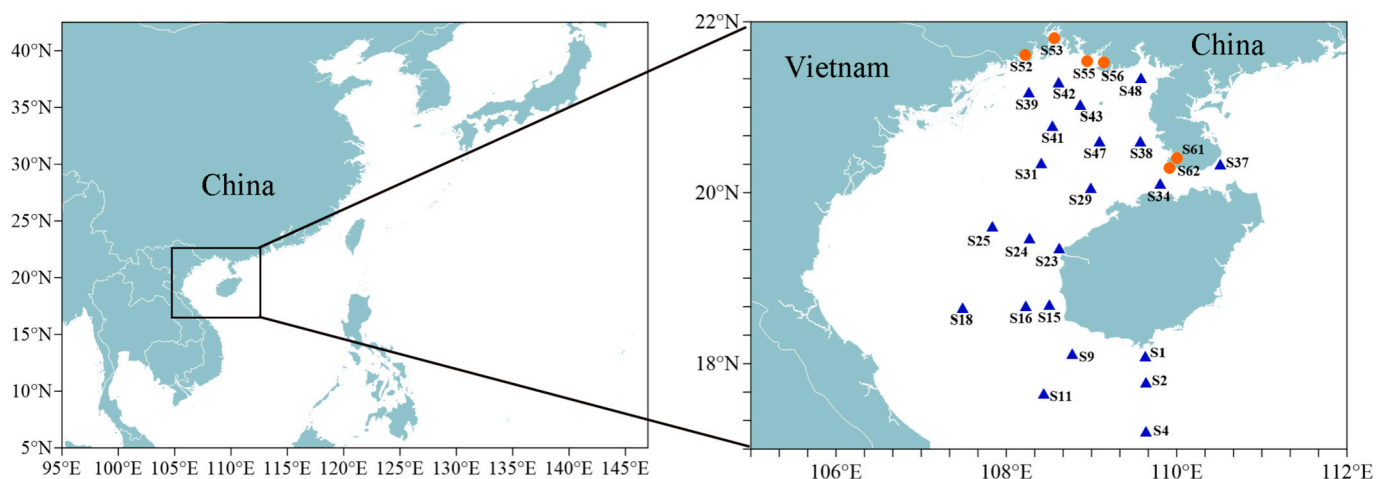
Sampling was conducted at 62 stations in the Beibu Gulf of the South China Sea during summer (August to September of 2021) and winter (December 2021 to January 2022) (Zheng et al., 2025). Lugol-fixed samples were collected by filtering 500–1000 mL of seawater on site, followed by fixation with Lugol's solution (final concentration of 1 %), and stored at 4 °C. In the laboratory, samples were allowed to settle for 48 h, and the supernatant was carefully siphoned off to concentrate the remaining material.

DNA samples were simultaneously collected for metabarcoding analysis. For each station, 5 mL of the Lugol-fixed sample was used for IFCB image acquisition. The acquired images were classified using the ViT model, with the category corresponding to the highest confidence assigned as the predicted class. Images with a confidence level below 90 % were classified as “Other”. In addition, 28 stations (6 coastal and 22 offshore) were selected for microscopic examination (Fig. 9).

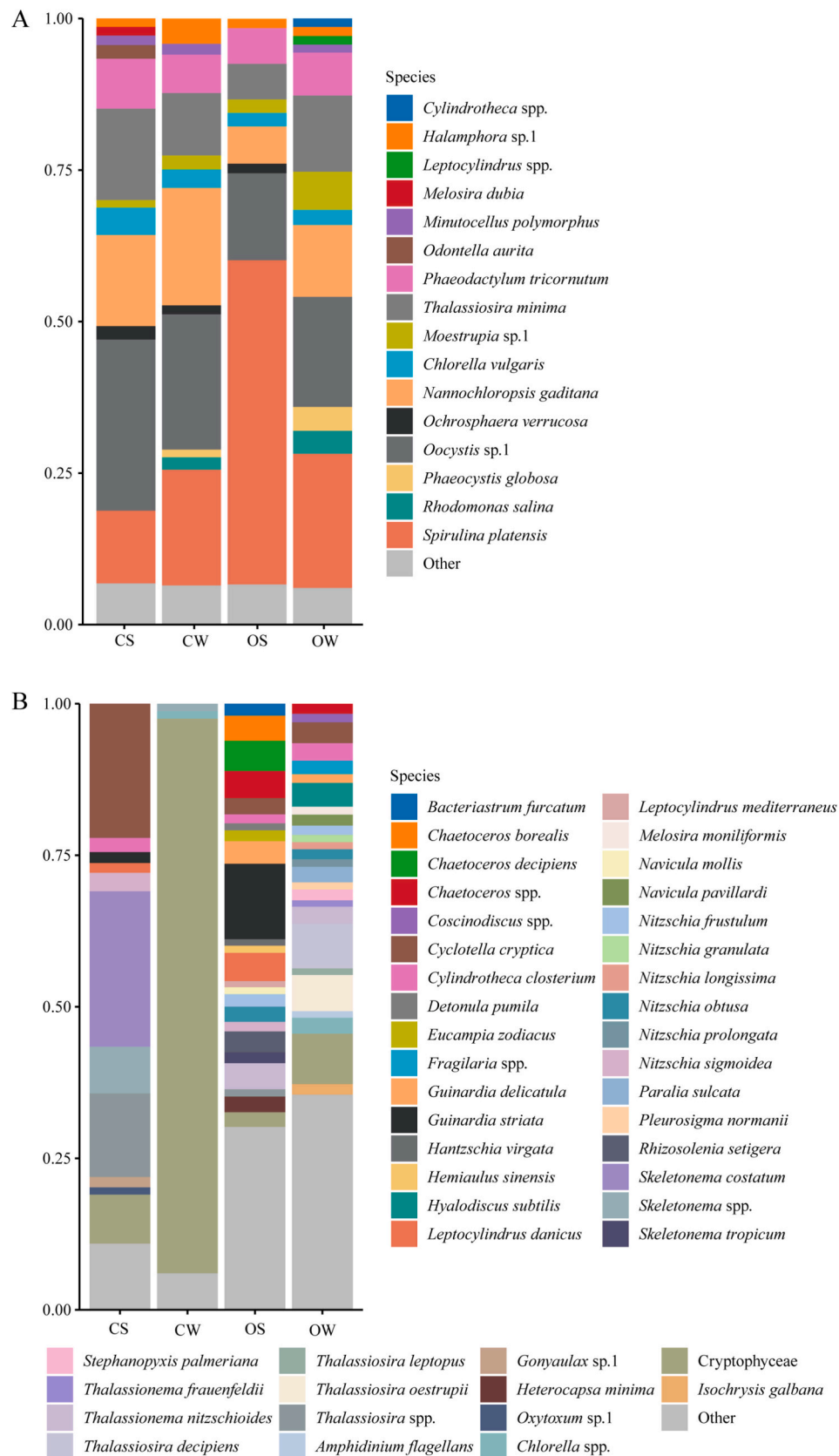
The AI-based classification results revealed that *Spirulina platensis*, a representative species of Cyanophyceae, exhibited the highest relative abundance in offshore waters, particularly in summer, accounting for over 50 %. In contrast, in coastal waters, *Oocystis* sp.1 (Trebouxiphyceae) was the most dominant, followed by *Nannochloropsis gaditana*. Among diatoms, *Phaeodactylum tricornutum* and *Thalassiosira minima* showed the highest relative abundance. Notably, *T. minima* exceeded 10 % in both coastal areas (summer and winter) and offshore areas (winter). Among dinoflagellates, *Moestrupia* sp.1 was the only species with a relative abundance above 6 % in offshore areas during winter (Fig. 10A).

Microscopic analysis identified a total of 241 genera or species, including 168 diatoms, 64 dinoflagellates, and 9 other microalgal taxa. At the species level, the dominant taxa in coastal waters during summer were *Skeletonema costatum*, while Cryptophyceae dominated coastal waters in winter, followed by *Skeletonema* spp. In offshore waters, *Guinardia striata* was dominant in summer, whereas Cryptophyceae and *Thalassiosira decipiens* were most abundant in winter. The relative abundance of individual dinoflagellate species was generally low (< 3 %) (Fig. 10B). A *t*-test of microalgal abundance indicated a statistically significant difference ( $p < 0.05$ ) between the community structures obtained by the two approaches. The model-based method detected more Cyanobacteria and Chlorophyta, while microscopy tended to record diatoms more effectively due to their distinct morphology, whereas smaller-celled taxa were more likely to be overlooked (Esenkulova et al., 2020).

Metabarcoding analysis revealed the presence of 261 microalgal



**Fig. 9.** Microscopic examination station sites in the Beibu Gulf. (Orange dots indicate coastal area, blue triangles indicate offshore area.)



**Fig. 10.** Relative abundance of microalgae at species level in the Beibu Gulf. A: AI classification; B: Microscopic analysis. (CS: stations in coastal area of summer time; CW: stations in coastal area of winter time; OS: stations in offshore area of summer time; OW: stations in offshore area of winter time; Other: sum of species with relative abundance less than 1 %.)

species in total, spanning 22 taxonomic classes. The identified species composition included 95 diatoms, 108 dinoflagellates, and 58 other algal taxa (including unclassified units). The highest species richness (217 species) was recorded in offshore waters during summer. Chlorophyta exhibited high relative abundance, particularly in offshore waters during winter (close to 50 %), while Cryptophyceae showed high abundance in winter, especially in coastal areas (approximately 30 %). In coastal waters during summer, *Chaetoceros tenuissimus* reached a relative abundance of 20 % (Zheng et al., 2025). Compared to the AI-based detection approach, metabarcoding identified a greater number of species and higher relative abundance of dinoflagellates, owing to the more comprehensive molecular reference databases developed over years of research. In contrast, the AI model is currently limited by the smaller number of species in its image database. A Wilcoxon rank-sum test revealed significant differences ( $p < 0.05$ ) in community composition between the two approaches.

The AI model currently includes 140 microalgal genera or species, which is fewer than those identified by microscopy or metabarcoding. To further enhance model performance, expanding the database to include more taxa and diverse images is essential (Li and Shang, 2024). Nevertheless, the IFCB-based AI classification approach offers substantial advantages in high-throughput image analysis, drastically reducing processing time and improving analytical efficiency. Moreover, it minimizes dependence on individual expertise required for microscopy and eliminates the complex preprocessing steps needed for high-throughput sequencing, thereby lowering both labor cost and operational barriers.

## 5. Conclusion

In this study, we developed a comprehensive microalgae image database and applied advanced artificial intelligence algorithms to enhance the accuracy of microalgae classification, with a particular focus on HAB species. The database comprises 142 categories, including 140 microalgae species and 106,606 images, providing a robust foundation for training and testing deep learning models. We selected three models—DenseNet, EfficientNet, and Vision Transformer (ViT)—and evaluated their performance. The results showed that ViT achieved the highest F1 score of 0.9205, followed by EfficientNet at 0.9169, and DenseNet at 0.9048. The study revealed that larger and more diverse datasets significantly improve model performance, with the best results obtained when each category contained more than 300 images. Furthermore, the trained models were applied to field-collected seawater samples, and the classification results were compared with those obtained from microscopic examination and metabarcoding analyses, demonstrating the practical applicability of the approach. Additionally, the application of AI technology in microalgae classification offers a reliable basis for early warning and rapid response to HABs, which can mitigate ecological and economic impacts. Our findings underscore the potential of AI in enhancing the speed and accuracy of microalgae identification, contributing to better management and protection of marine ecosystems.

## CRediT authorship contribution statement

**Junjie Zheng:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zihan Sun:** Writing – review & editing, Validation, Software, Investigation, Formal analysis, Data curation. **Ruoyu Guo:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Ruifang Wang:** Validation, Resources, Methodology, Formal analysis. **Dian Jin:** Visualization, Resources, Data curation. **Jiarong Hu:** Methodology, Investigation. **Xiaogang Xing:** Resources. **Mengmeng Tong:** Resources. **Pengbin Wang:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal

analysis, Data curation, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2025.119170>.

## Data availability

The image dataset, trained models, and relevant code used in this study are available from the corresponding author upon reasonable request.

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